

ASHUTOSH ROY

AI/ML Researcher · NLP & Multimodal Systems · Published at IEEE RECCAP 2026 & IIT Delhi

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SUMMARY

AI/ML researcher and engineer with **2 published papers** (RECCAP 2026, IIT Palakkad; co-authored with IIT Delhi). Hands-on experience building large-scale NLP, multimodal AI, RAG, and speech processing systems. Skilled in transformer fine-tuning, hybrid retrieval pipelines, and end-to-end ML system deployment. Proficient in **Python, C++**, PyTorch, TensorFlow, and LangChain. Open-source contributor with production projects serving real users. Seeking to contribute to Google's research agenda on real-world, large-scale AI problems.

EXPERIENCE

Research Intern

Jan 2025 – Jun 2025

Indian Institute of Technology (IIT) Delhi

New Delhi, India

- Built an OCR-powered document understanding pipeline for archaeological datasets; benchmarked Tesseract, Marker, and Gemini OCR — selecting optimal strategy per document class to improve extraction accuracy.
- Designed hybrid retrieval (dense: FAISS/Pinecone + sparse: Apache Solr) for high-precision semantic + lexical document search over thousands of historical records.
- Deployed scalable **FastAPI** services for real-time LLM-based querying; integrated with frontend for sub-second retrieval responses.
- Co-authored **SARCH: Multimodal Search for Archaeological Archives** with IIT Delhi collaborators — submitted for peer review. arxiv.org/abs/2511.05667

AI/ML Engineer Intern

Oct 2024 – Nov 2024

MitoVoid AI

Remote

- Fine-tuned transformer-based NLP models (Mistral/LLaMA variants) for healthcare applications — medical record summarization and symptom-driven conversational AI.
- Reduced patient query resolution time by ~**40%** by deploying an AI chatbot for telemedicine with optimized intent understanding and response generation.
- Implemented GenAI-powered NLP workflows using LLMs and retrieval-augmented pipelines for intelligent healthcare automation and conversational AI systems.
- Built end-to-end ML pipelines: data preprocessing → model inference → API deployment for real-time healthcare applications.

PUBLICATIONS & RESEARCH

SARCH: Multimodal Search for Archaeological Archives | Co-authored with IIT Delhi (Prof. Maya Ramanath)

2025

- Proposed a multimodal retrieval system combining OCR (Surya), image classification, and hybrid search (FAISS + Apache Solr) for large-scale scanned archaeological archives.
- Benchmarked multiple OCR engines (Tesseract, Marker, Gemini OCR) across diverse historical document layouts; achieved significant gains in extraction accuracy.

Multitask Speech Emotion & Stress Detection (Wav2Vec2-BiLSTM) | Accepted — RECCAP 2026, IIT Palakkad (IEEE)

2026

- Proposed a self-supervised Wav2Vec2 + BiLSTM multitask architecture for simultaneous 5-class emotion classification and continuous stress regression; achieved **85.9% accuracy**, **0.856 F1**, RMSE **0.1268**.
- Evaluated across 3 corpora (RAVDESS, TESS, SAVEE) with unified preprocessing pipeline (VAD, silence removal, log-Mel spectrograms) and data augmentation.

PROJECTS

Archaeological Full-Stack Search Engine | Python, Surya OCR, Apache Solr, FAISS, FastAPI, C++, NLP

Jan 2025 – Jun 2025

- Built an end-to-end search engine for archaeological archives at IIT Delhi, processing scanned manuscripts, books, maps, and historical records.
- Developed OCR and document understanding pipelines using Surya OCR to convert complex multi-layout documents into structured Markdown and JSON formats.
- Implemented metadata extraction and semantic indexing using Apache Solr and FAISS, enabling sub-second keyword and vector-based retrieval.
- Designed multimodal search workflows supporting text and image retrieval across large archaeological collections.
- Deployed the system using FastAPI for real-time search and document exploration.

Industrial Defect Detection and Quality Inspection System | YOLOv8, PyTorch, OpenCV, CUDA, Python

Feb 2025 – Apr 2025

- Developed a real-time computer vision system for automated quality inspection in manufacturing environments using YOLOv8.
- Collected and annotated industrial datasets containing defects such as scratches, dents, cracks, missing components, and surface anomalies.
- Fine-tuned YOLOv8 object detection models on custom datasets, achieving high detection accuracy while maintaining low-latency inference.
- Implemented image augmentation and preprocessing pipelines to improve robustness under varying factory lighting and camera conditions.
- Built automated defect reporting, visualization, and analytics modules to support production-line quality monitoring.

Speech Emotion & Stress Detection | PyTorch, Wav2Vec2, BiLSTM, Hugging Face, Python

RECCAP 2026

- Developed a multi-task deep learning model using **Wav2Vec2** and **BiLSTM** for simultaneous speech emotion classification and stress prediction.
- Preprocessed and combined **RAVDESS**, **TESS**, and **SAVEE** datasets with feature engineering, silence removal, and data augmentation to improve model performance.
- Achieved **85.9% accuracy**, **0.856 F1-score**, and **0.1268 RMSE** through hyperparameter tuning and comprehensive model evaluation.
- Evaluated the model using confusion matrix, precision, recall, F1-score, and RMSE, outperforming baseline models on emotion recognition tasks.
- Research accepted at **RECCAP 2026 (IEEE)**, demonstrating improved speech emotion and stress detection for healthcare applications.

Exam-Helper RAG System | Python, LLaMA 3, FAISS, Pinecone, LangChain, Streamlit

Dec 2024 – Apr 2025

- Built a Retrieval-Augmented Generation (RAG) system powered by LLaMA 3 for contextual question answering over custom knowledge bases.
- Implemented hybrid retrieval pipelines using FAISS and Pinecone for efficient semantic search and context retrieval.
- Developed multimodal ingestion pipelines supporting PDFs, images, and YouTube transcripts.
- Reduced hallucinations and improved answer relevance through optimized chunking, embedding, and retrieval strategies.
- Deployed an interactive Streamlit application for end-user document querying and knowledge discovery.

Medical Chatbot and LLM Fine-Tuning | Mistral 7B, QLoRA, Transformers, Tesseract OCR, Python

Mar 2024 – Apr 2024

- Fine-tuned Mistral 7B using QLoRA with 4-bit quantization for healthcare-focused conversational AI applications.
- Built OCR pipelines for extracting structured information from unstructured medical reports and prescriptions.
- Implemented efficient parameter-efficient fine-tuning techniques, significantly reducing GPU memory requirements.
- Developed inference pipelines for medical record summarization, question answering, and symptom-based assistance.
- Evaluated model performance using domain-specific benchmarks and qualitative human assessment.

OPEN SOURCE CONTRIBUTIONS

InterviewPrep — Open-Source AI/ML Interview Platform | Astro, JavaScript, HTML/CSS, SEO

May 2026 – Jun 2026

- Developed and open-sourced a comprehensive interview prep platform with **700+ curated questions** spanning Python, ML, Deep Learning, NLP, LLMs, RAG, Computer Vision, SQL, and HR.
- Features company-specific tracks (FAANG, OpenAI, NVIDIA, Indian IT), a 12-week structured roadmap, interactive flashcards, and role-specific mock interview simulations.
- Fully responsive static site with light/dark mode, optimized schema markup, and **95+ Lighthouse performance score**; designed for large-scale community use.

Skills-for-Agents — Composable Instruction Sets for Multi-Task AI Agents | Python, LLM Agents, SIP Protocol

May 2026

- Built a production-ready ecosystem of modular, conflict-free instruction sets for multi-task LLM agents; each skill owns exactly one domain (voice, density, craft, process, or content).
- Designed a custom **Skills Interoperability Protocol (SIP)** to completely prevent instruction conflicts during concurrent multi-agent execution, enabling scalable and reliable agent workflows.

BulkyMail — Automated Bulk Email Campaign Tool | Python, Streamlit, SMTP/API

Dec 2024

- Built a Streamlit web app for automated, personalized bulk email campaigns; enabled users to send **100+ personalized emails in minutes** with dynamic templating (name, role, company).
- Contributed to successful internship conversions for multiple users by improving outreach response rates through personalized messaging at scale.

TECHNICAL SKILLS

AI/ML & Research: LLMs, GenAI, RAG, NLP, Transformer Fine-Tuning (QLoRA/LoRA), Self-Supervised Learning, Multimodal AI, Speech Processing, Computer Vision, Dense Retrieval, Foundation Models, Evaluation & Benchmarking

Languages: Python, C++, SQL, JavaScript

Libraries & Frameworks: PyTorch, Hugging Face Transformers, LangChain, TensorFlow, Scikit-learn, Pandas, NumPy, FastAPI, Streamlit, OpenCV

Databases & Search: FAISS, Pinecone, Apache Solr, MySQL

SDLC & Engineering: Git, Docker, Linux, REST API Design, MLOps, Model Deployment, Agile workflows, CI/CD pipelines

Tools: Jupyter Notebook, Google Colab, VS Code, OpenAI GPT, Claude, Power BI, MATLAB

EDUCATION

Chhattisgarh Swami Vivekananda Technical University (CSVTU)

B.Tech (Hons) in Computer Science and Engineering (Artificial Intelligence)

2022 – 2026

Bhilai, Chhattisgarh

St. Columbus School

Higher Secondary (Class XII - Science PCMB) - 87%

2020 – 2021

Faridabad, Haryana